



ST BENEDICT'S

SUBJECT	Physical Science: Paper 1	DATE	16 November 2020
GRADE	11	MARKS	200 Marks
EXAMINER	Mr McCoy	MODERATOR	Mr du Toit
NAME		DURATION	3 hours
CLASS			

LOW ORDER 50%

HIGH ORDER 50%

INSTRUCTIONS

1. This question paper consists of 13 pages, an Answer Sheet of 2 pages and a Data Sheet of 2 pages. Please make sure that your question paper is complete.
2. Answer ALL the questions.
3. Read the questions carefully.
4. Use the data and formulae whenever necessary.
5. **Start each question on a new page.**
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in all answers.
8. Express all answers to TWO decimal places unless told otherwise.
9. It is in your own interest to write legibly and to present your work neatly.

Question	1	2	3	4	5	6	7	8	9	TOTAL
Total	20	25	18	17	32	35	19	22	12	200

QUESTION 1

20 MARKS

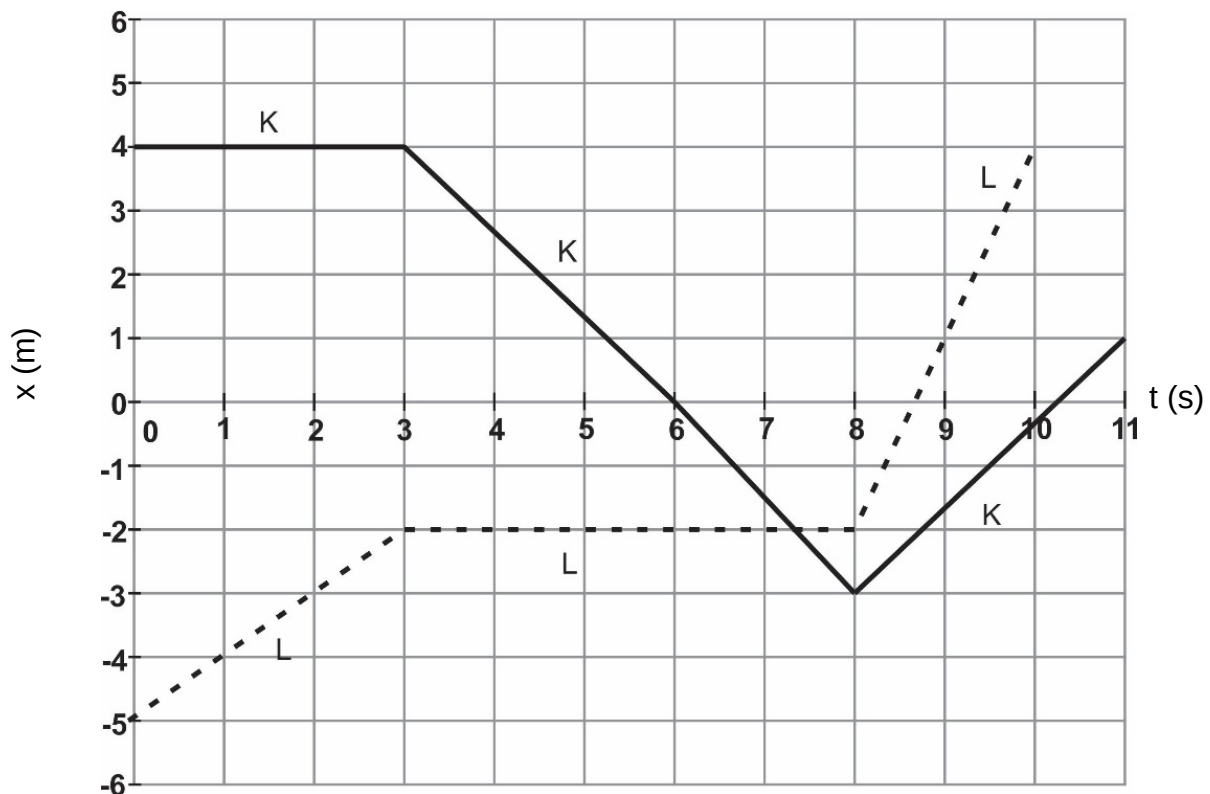
Answer the following multiple-choice questions on your separate ANSWER SHEET. Make a cross (X) in the box corresponding to the letter which you consider to be the most correct.

1.1 Which of the following pairs of quantities are BOTH vector quantities?

- A direction and displacement
- B acceleration and position
- C charge and gravitational field strength
- D voltage and acceleration

(2)

1.2 The graph shows the position (x) of two soccer players, K and L, moving along the same horizontal straight line.



At $t = 3$ s, what is the distance between K and L?

- A -2 m
- B 2 m
- C 4 m
- D 6 m

(2)

1.3 Which one of the following statements is TRUE of a body that accelerates uniformly?

- A Its rate of change of position with time remains constant.
- B Its position changes by the same amount in equal time intervals.
- C Its velocity increases by the same amount in equal time intervals.
- D Its velocity increases by increasing amounts in equal time intervals.

(2)

1.4 According to Newton's Second Law of Motion, the acceleration of an object is....

- A independent of its mass.
- B always equal to its mass.
- C directly proportional to its mass.
- D inversely proportional to its mass.

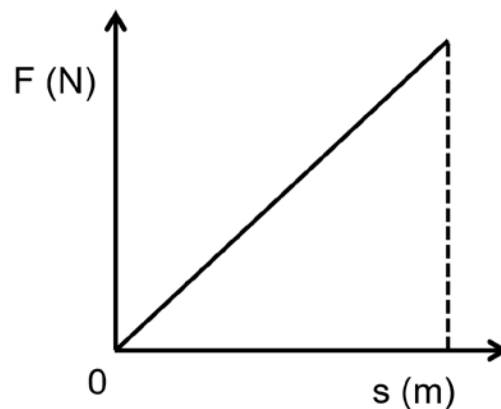
(2)

1.5 A force of 58 N is applied to a 4 kg box to slide it 10 m along a horizontal, rough surface with a friction force of 2 N. How much work is done to overcome gravity?

- A 0 N.m
- B 39,2 N.m
- C 392 N.m
- D 560 N.m

(2)

1.6 Consider the following graph below.

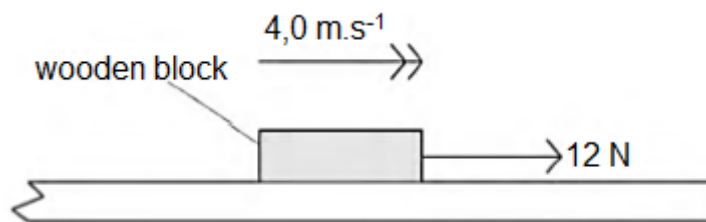


What physical quantity does the area under the graph represent?

- A work
- B power
- C acceleration
- D velocity

(2)

- 1.7 A wooden block of mass 0,6 kg is accelerated along a rough horizontal surface at 4 m.s^{-2} by a pulling force of 12 N.

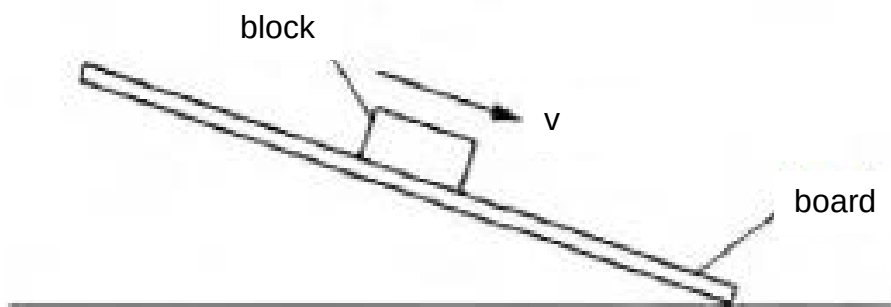


What is the magnitude of the frictional force acting on the block?

- A 2,4 N
- B 9,6 N
- C 14,0 N
- D 16,0 N

(2)

- 1.8 A wooden block rests on a rough board. The end of the board is then raised until the block slides down the plane of the board at a constant velocity, v .

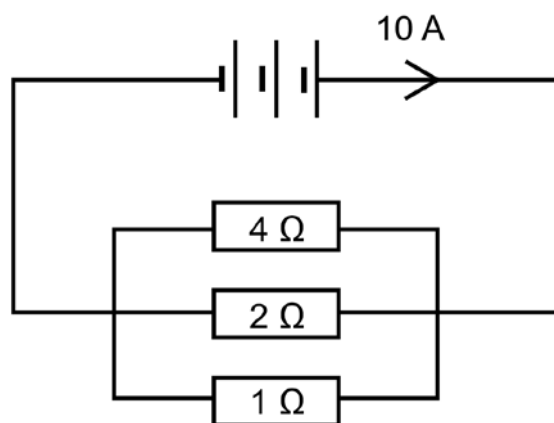


Which row correctly describes the forces acting on the block while it is sliding?

	Frictional force	Resultant force
A	Down the plane	Down the plane
B	Down the plane	Zero
C	Up the plane	Down the plane
D	Up the plane	Zero

(2)

1.9 The current in a cell is 10 A has shown.

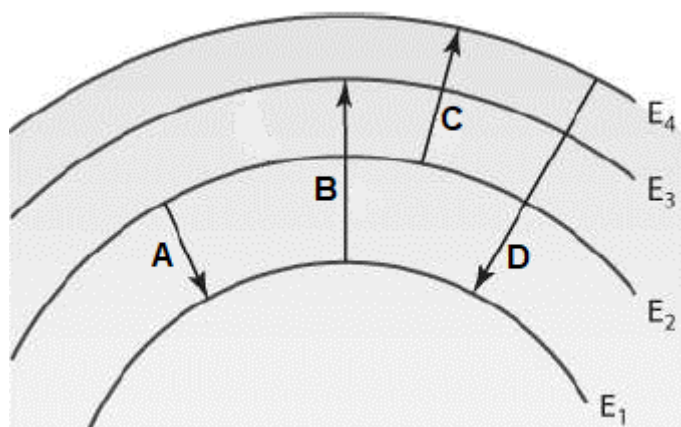


What is the current in the 2 Ω resistor?

- A 0,35 A
- B 2,86 A
- C 3,50 A
- D 7,14 A

(2)

1.10 The diagram below shows possible transitions of electrons between energy levels of an element. Which transition would produce the line of SHORTEST WAVELENGTH on the emission spectrum of the element?



(2)

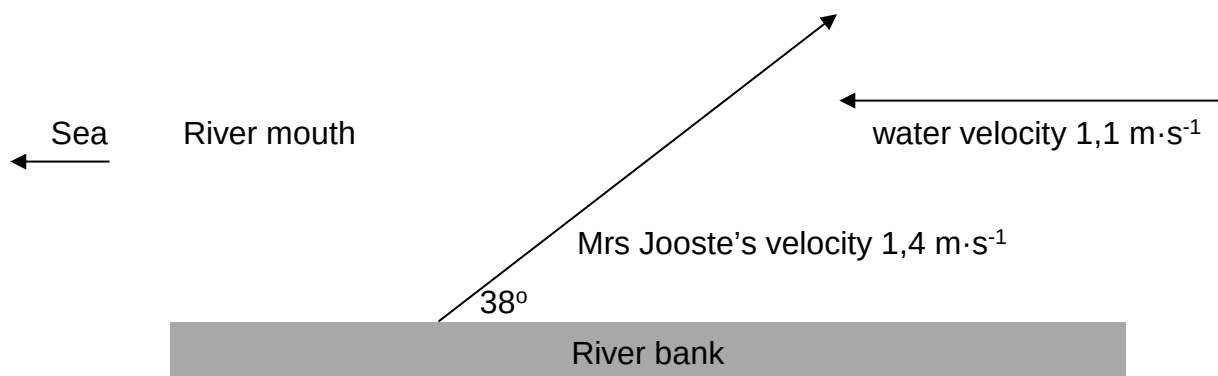
QUESTION 2

25 MARKS

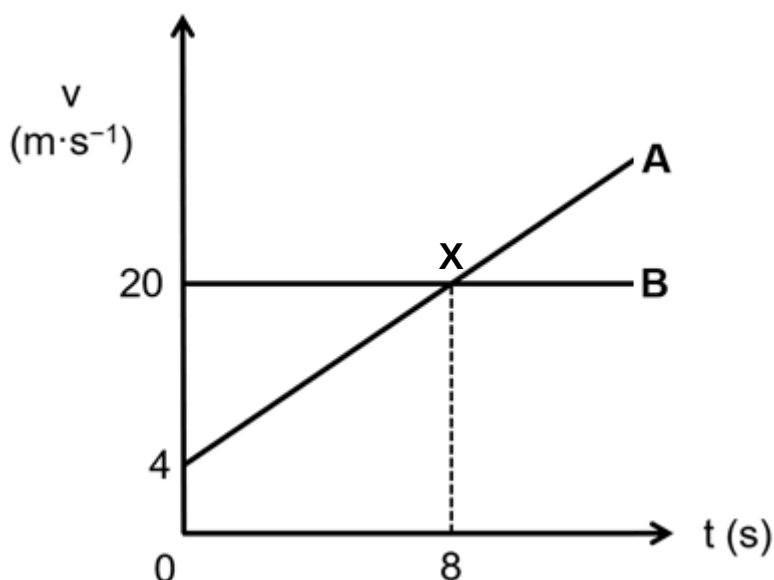
- 2.1 Mrs Jooste went for a run on the beach at Port St John's. She must cross the *Umzimvubu* river mouth before it is completely dark. She has 5 minutes before the sun is completely down.

Refer to the diagram of the scenario below:

The river is flowing out to sea and Mrs Jooste jumps into the river at an angle of 38° upstream to the river bank because she wants to go directly across the river which is 89 m wide.



- 2.1.1 Draw a labelled right-angled vector triangle to represent Mrs Jooste's swim across the river mouth. (4)
- 2.1.2 Define *resultant vector*. (2)
- 2.1.3 Calculate the resultant velocity of Mrs Jooste. (4)
- 2.1.4 Did Mrs Jooste make it across the river mouth before sundown? Show all necessary calculations to support your answer. (5)
- 2.2 The following velocity-time graph shows the motion of two cars, A and B.

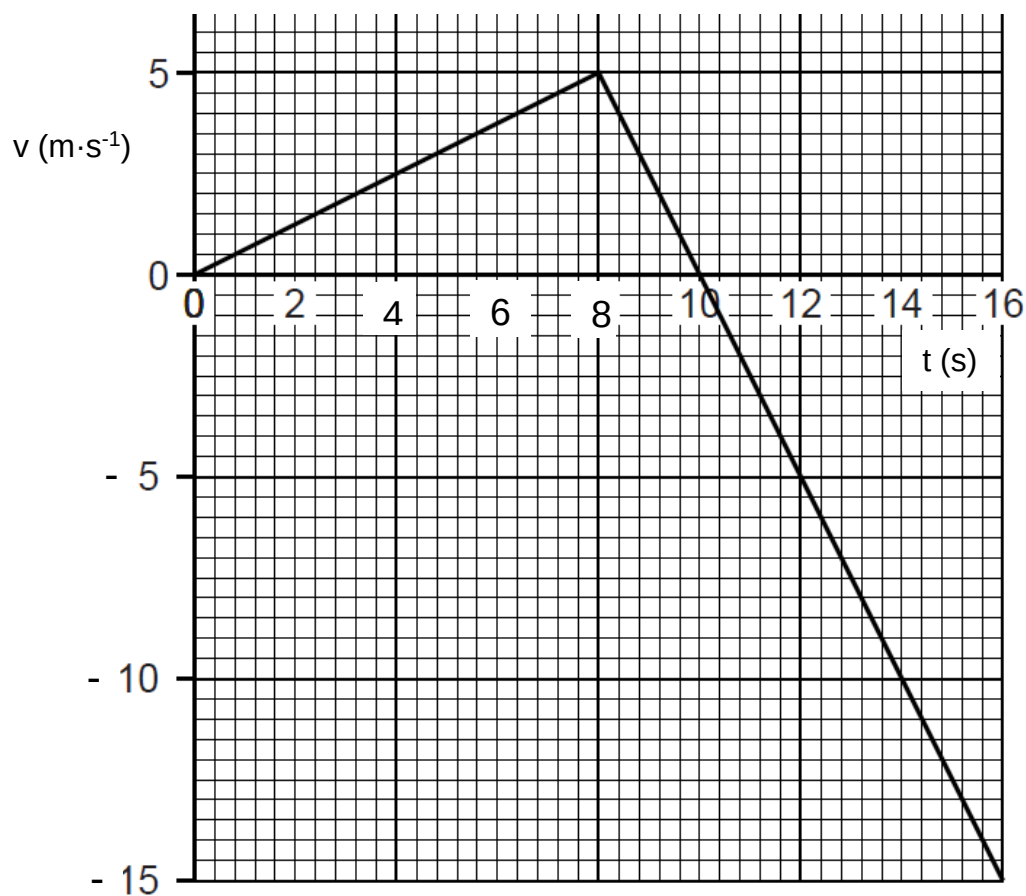


- 2.2.1 Define *velocity*. (2)
- 2.2.2 The two graphs intersect at point X. What can you conclude about the cars at this point? (1)
- 2.2.3 At what distance after time zero will car A catch up to car B? (7)

QUESTION 3

18 MARKS

The variation with time t of the velocity v of a ball is shown in the graph below.



The ball moves forwards in a straight line from a point at $t = 0$ s.

3.1 Use the graph to calculate:

3.1.1 the magnitude of the displacement at $t = 10$ s. (4)

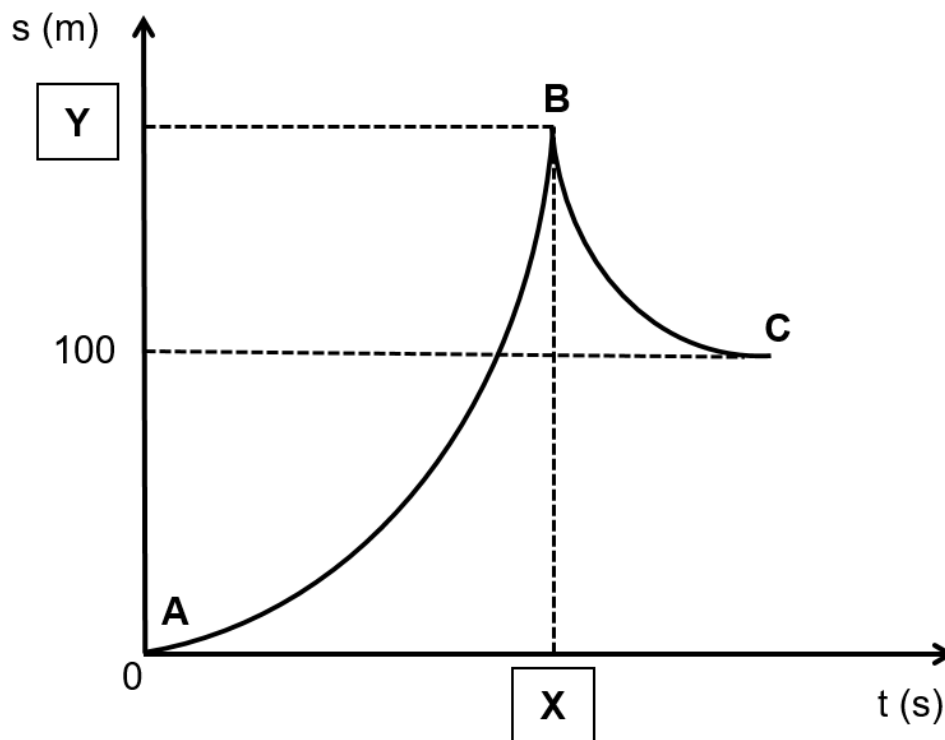
3.1.2 the acceleration at 10 s. (4)

3.2 Use your answers above to determine the time from $t = 0$ s for the ball to return to its starting point. (5)

3.3 Sketch the position vs. time graph which corresponds to the motion of the Ball from 0 – 16 s. Include all known values. (5)

QUESTION 4**17 MARKS**

A child throws a ball vertically downwards from a window of a high building with an initial velocity of $15 \text{ m}\cdot\text{s}^{-1}$ at point A. It strikes the ground travelling at $55 \text{ m}\cdot\text{s}^{-1}$ at point B and bounces up off the ground to an unknown maximum height at point C as shown on the following position time graph below. Ignore the effects of air resistance.



- 4.1 Describe the motion of the ball from A to B. (2)
- 4.2 Calculate the value of X on the graph. (3)
- 4.3 Calculate the value of Y on the graph. (3)
- 4.4 Hence, determine the maximum height that the ball reaches above the ground after the bounce. (2)
- 4.5 Describe the feature of the graph that indicates that, at point C, the ball is at maximum height above the ground after the bounce. (2)
- 4.6 ON THE ANSWER SHEET, draw a velocity versus time sketch graph for the motion of the ball from A to C taking down as positive. You must include only known values on this graph and the relative sizes of the velocity values on your graph must be correct. (5)

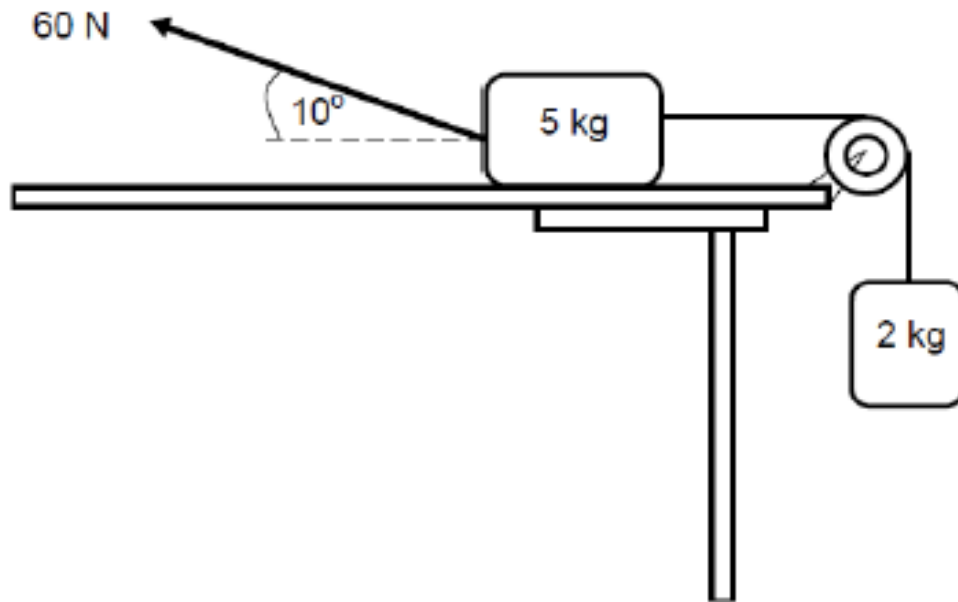
QUESTION 5

32 MARKS

5.1 A 5 kg block, resting on a rough horizontal table, is connected by a light inextensible string passing over a light frictionless pulley to another block of mass 2 kg.

The 2 kg block hangs vertically as shown in the diagram below.

A force of 60 N is applied to the 5 kg block at an angle of 10° to the horizontal, causing the 5 kg block to accelerate to the left.



The coefficient of kinetic friction between the 5 kg block and the surface of the table is 0,5. Ignore the effects of air resistance.

- 5.1.1 Draw a labelled free-body diagram showing ALL the forces acting on the 5 kg block. (5)
- 5.1.2 State the name and direction force of the reaction force to the weight of the 5 kg block. (3)
- 5.1.3 Calculate the magnitude of:
 - a) the vertical component of the 60 N force. (2)
 - b) the horizontal component of the 60 N force. (2)
- 5.1.4 Define the *normal force*. (2)
- 5.1.5 Calculate the magnitude of the normal force acting on the 5 kg block. (3)
- 5.1.6 Define *frictional force due to a surface*. (2)
- 5.1.7 Calculate the magnitude of the force of friction acting on the 5 kg block. (2)
- 5.1.8 Calculate the magnitude of the acceleration of the system and the tension in the string connecting the two blocks. (6)

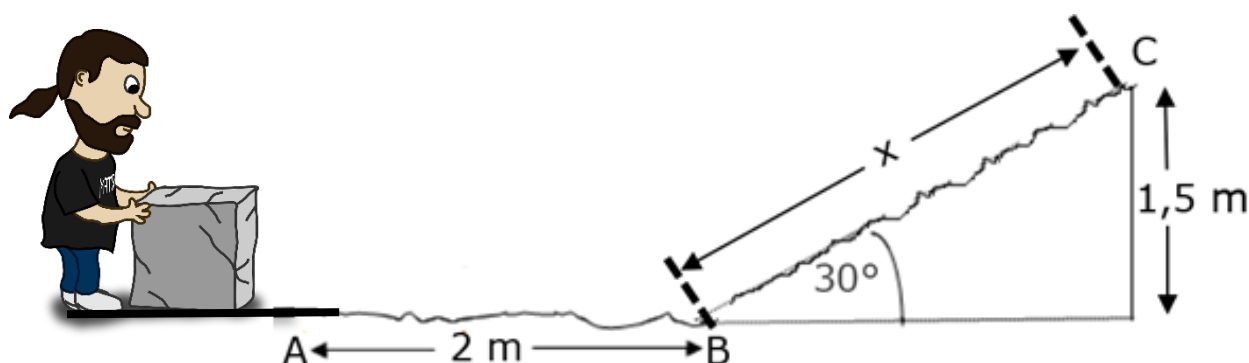
5.2 Studies suggest that wearing a seat belt is the best way to prevent injury while in a motor vehicle. The Journal of the Royal Society of Medicine revealed that seat belt laws have reduced injury rates by 40 percent and also reduced the severity of injuries by 35 percent.

Explain how a seat belt reduces injuries on a passenger during an accident, by naming and referring to a relevant physics law. (5)

QUESTION 6

35 MARKS

6.1 A man pushes a granite block of mass 200 kg up a slope.



The velocity of the block at point A is 5 m.s^{-1} to the right. From point A to B, a rougher 2 m section, the block passes point B at 3 m.s^{-1} . The block then continues up the rough slope, of vertical height 1,5 m and a slope length of x. The slope exerts a constant average frictional force on the block and it stops at point C.

6.1.1 Define *work*. (2)

6.1.2 If the man applies a force of 300 N over the 2 m flat stretch, calculate the friction between the block and slope **using energy principles**. (6)

6.1.3 The block continues up the rough slope until it comes to rest after covering a distance x.

a) Calculate the distance x up the slope. (2)

b) State the *work-energy theorem*. (2)

c) Calculate the change in kinetic energy from point B to point C of the block. (4)

d) Calculate the magnitude of the applied force on the block if there is a constant frictional force of 400 N. (6)

e) Calculate how much work is done by the man from point B to point C? (3)

6.2 A 1,4 kg toy rocket is launched straight upwards using 2000 J of chemical energy.

After 1,16 s the rocket has gained 343 J of gravitational potential energy and 1294,3 J of kinetic energy.

6.2.1 Determine the energy lost to friction in the motor and through the air. (4)

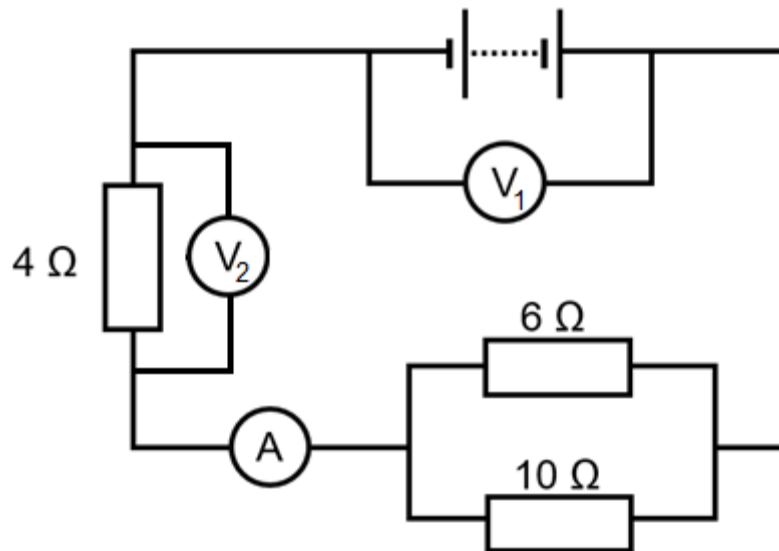
6.2.2 Calculate the power of the rocket. (3)

6.2.3 Determine the efficiency of the rocket. (3)

QUESTION 7

19 MARKS

A circuit that consists of a battery with an emf of 23,25 V, three resistors, an ammeter and 2 voltmeters are set up as shown in the diagram below.



7.1 Define *resistance*. (2)

7.2 Calculate the effective resistance of the circuit. (3)

The ammeter measures a current of 3 A with the circuit connected as shown.

7.3 Determine the current through the $6\ \Omega$ resistor. (3)

7.4 Determine the power of the $4\ \Omega$ resistor. (3)

Sometimes resistors burn out while in operation

7.5 Describe what would happen to the reading on the voltmeter 2 if the $6\ \Omega$ resistor burnt out. Use a suitable equation in your answer. (4)

7.6 Define *emf*. (2)

7.7 Predict what the reading on the voltmeter 1 would be if the $4\ \Omega$ resistor burnt out. Briefly explain your answer. (2)

QUESTION 8

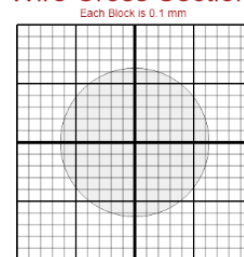
22 MARKS

Mr McCoy investigates the effect of gauge on the resistance of a metallic conducting wire.

He follows the method below:

1. Measure the diameter of the wire of length 3,6 m. Record in the table.
2. Determine the resistance of the wire. Record in the table.
3. Repeat steps 1 and 2 for four other diameters of the same wire.
4. Repeat 1 – 3. Record in the table.

Wire Cross Section



Average Gauge ($\times 10^{-3}$ m)	Average Resistance (Ω)	$\frac{1}{\text{gauge}^2}$ ($\times 10^6 \text{ m}^{-2}$)
0,48	19,53	4,34
0,72	8,49	Question 8.5
1,04	4,15	0,93
1,72	1,56	0,38
1.90	1,28	0,27

- 8.1 What is the independent variable in this investigation? (2)
- 8.2 List two controlled variables in this investigation, other than the type of material. (2)
- 8.3 Calculate the missing value in the third column. (2)
- 8.4 ON THE ANSWER SHEET plot a graph of Resistance vs $\frac{1}{\text{gauge}^2}$.
The first point has been plotted. Draw in a best fit line. (7)
- 8.5 Write out the relationship between gauge and resistance of a metal wire in words. (2)
- 8.6 Calculate the gradient of your graph. Give a unit. (4)
- 8.7 The following relationship is **known** regarding the resistance and cross sectional area of a wire, where ρ is known as the resistivity of the material, L is the length of the material used and A (gauge^2) is the cross sectional area. (3)

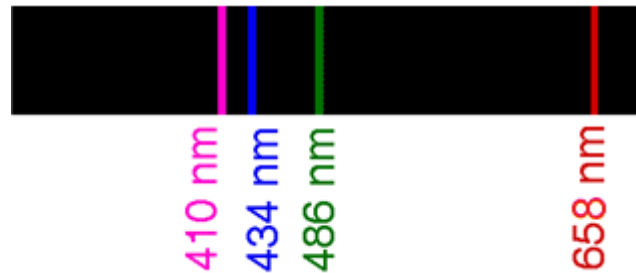
$$R = \rho \frac{L}{A}$$

Use this equation to determine the physical quantity or quantities represented by the gradient or your graph. (3)

QUESTION 9

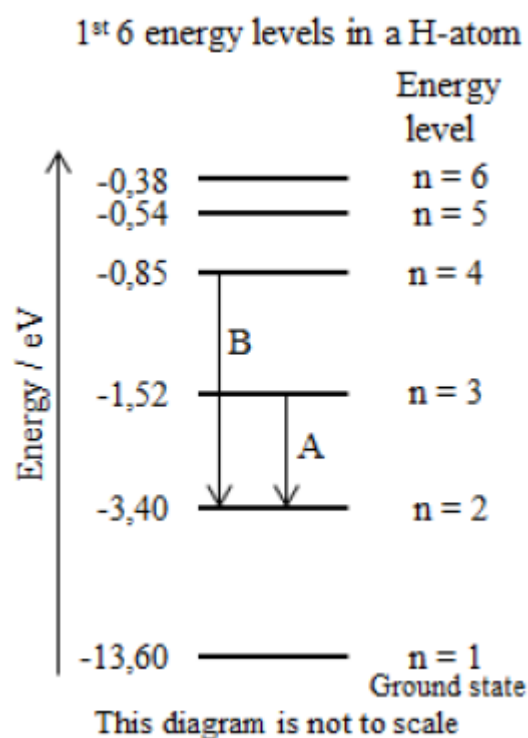
12 MARKS

The diagram shows the line-emission spectrum of hydrogen gas.



9.1 Explain, in bullet points, how the line-emission spectrum was produced. (5)

The diagram shows some of the energy levels of an isolated hydrogen atom. A and B represent transitions of electrons between energy levels. One of the lines produced by hydrogen gas is red with a wavelength of 658 nm.



9.2 Sketch a graph to show the relationship between energy and wavelength. (2)

9.3 Determine by calculation whether transition A or B is responsible for the red line (5)

END OF EXAMINATION



PHYSICAL SCIENCE PAPER 1

GRADE 11

16 NOVEMBER 2020

ST BENEDICT'S

ANSWER SHEET

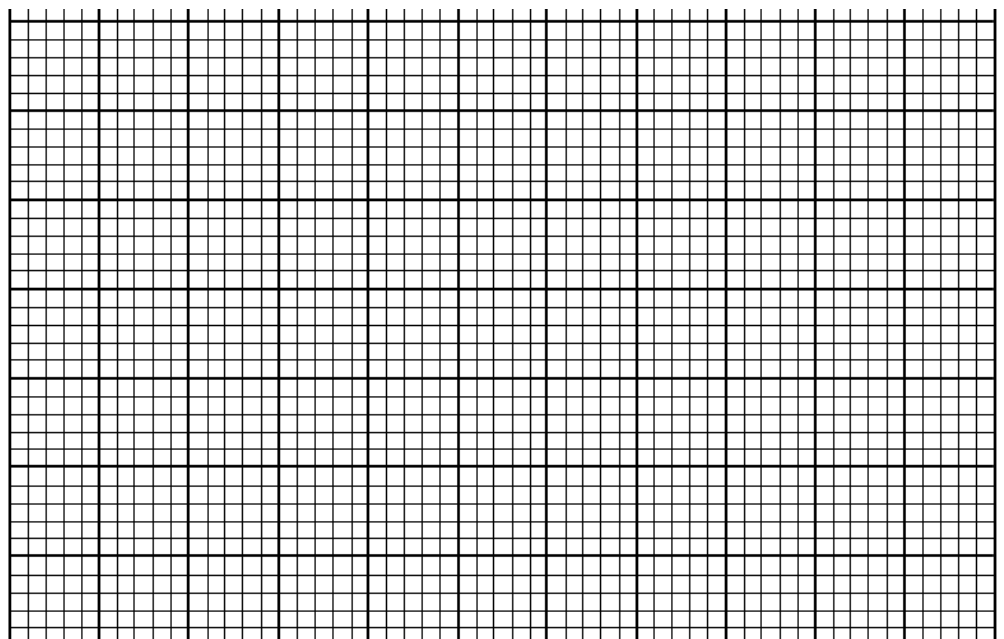
NAME: _____

QUESTION 1

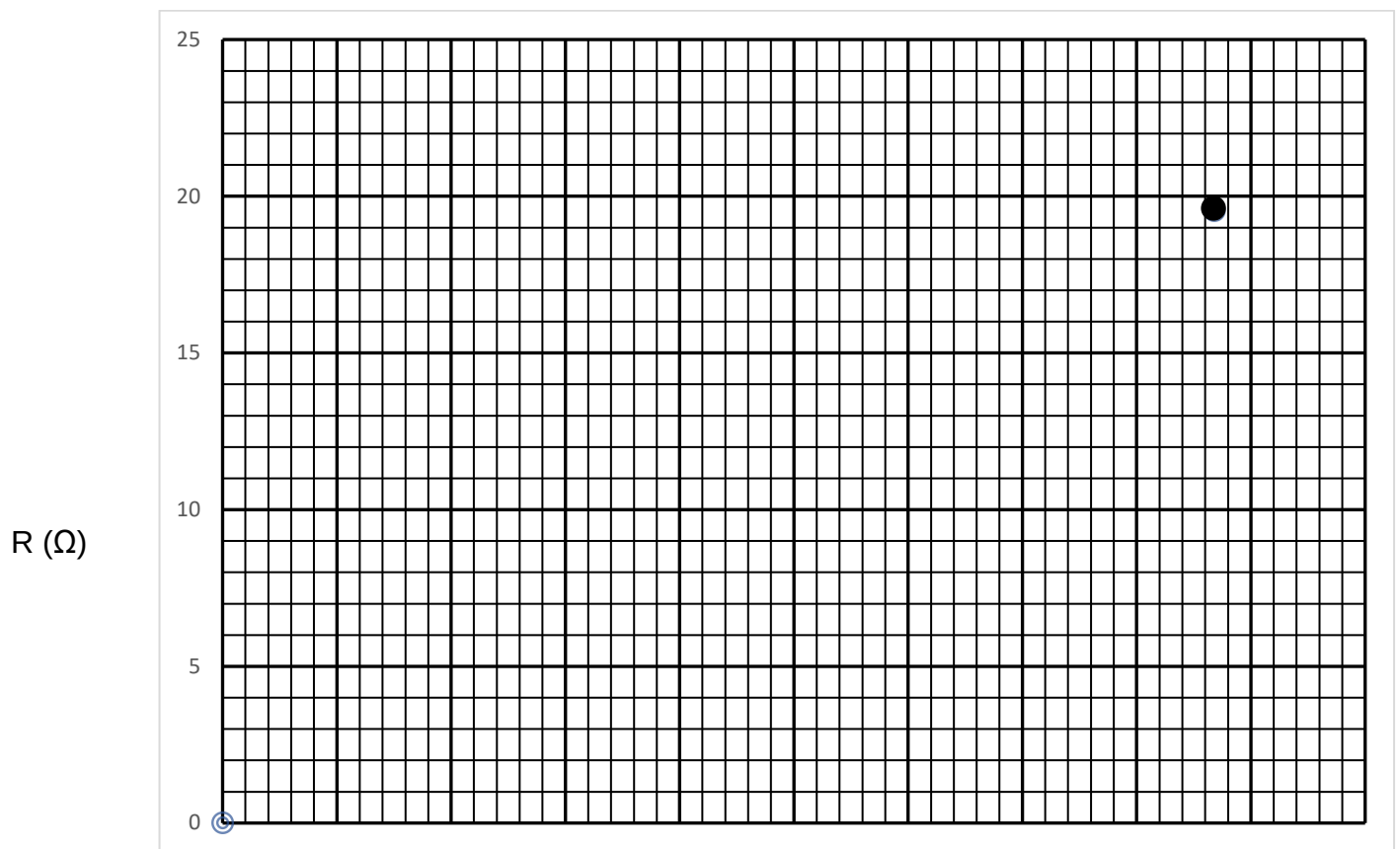
1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D
1.8	A	B	C	D
1.9	A	B	C	D
1.10	A	B	C	D

QUESTION 4.6

$v \text{ (m.s}^{-1}\text{)}$



QUESTION 8.4



**EXAMINATION DATA SHEET FOR THE PHYSICAL SCIENCES
(PHYSICS)**

TABLE 1 PHYSICAL CONSTANTS

NAME	SYMBOL	VALUE
Acceleration due to gravity	g	9,8 m·s ⁻²
Speed of light in a vacuum	c	3,0 × 10 ⁸ m·s ⁻¹
Universal gravitational constant	G	6,7 × 10 ⁻¹¹ N·m ² ·kg ⁻²
Coulomb's constant	k	9,0 × 10 ⁹ N·m ² ·C ⁻²
Magnitude of charge on electron	e	1,6 × 10 ⁻¹⁹ C
Mass of an electron	m _e	9,1 × 10 ⁻³¹ kg
Planck's constant	h	6,6 × 10 ⁻³⁴ J·s
1 electron volt	eV	1,6 × 10 ⁻¹⁹ J

TABLE 2 PHYSICS FORMULAE

MOTION

$v = u + at$ or $v_f = v_i + a\Delta t$	$s = \left(\frac{v+u}{2}\right)t$ or $\Delta x = \left(\frac{v_f+v_i}{2}\right)\Delta t$
$v^2 = u^2 + 2as$ or $v_f^2 = v_i^2 + 2a\Delta x$	$s = ut + \frac{1}{2}at^2$ or $\Delta x = v_i\Delta t + \frac{1}{2}a(\Delta t)^2$

FORCE AND MOMENTUM

$F_{net} = ma$	$F_{net} = \frac{\Delta p}{\Delta t}$ or $F_{net}\Delta t = m\Delta v$	$J = \Delta p = mv - mu$ or $J = \Delta p = mv_f - mv_i$
$p = mv$	$F_g = mg$	$F_{fs}^{max} = \mu_s F_N$ $F_{fk} = \mu_k F_N$

WORK, ENERGY AND POWER

$W = Fs$ or $W = F\Delta x$ or $W = F\Delta x \cos\theta$	$P = \frac{W}{t}$	$P = Fv$
$E_p = mgh$	$E_k = \frac{1}{2}mv^2$	$W_{net} = \Delta E_K$
efficiency = $\frac{\text{power}_{out}}{\text{power}_{in}} \times 100$		

GRAVITATIONAL AND ELECTRIC FIELDS

$F = G \frac{m_1 m_2}{r^2}$	$g = \frac{F}{m}$	$g = G \frac{M}{r^2}$
$F = k \frac{q_1 q_2}{r^2}$	$E = \frac{F}{q}$	$E = \frac{kQ}{r^2}$

ELECTRIC CIRCUITS

$I = \frac{q}{t}$	$V = \frac{W}{q}$
$R = \frac{V}{I}$	$emf = I(R_{ext} + r)$ or $emf = V_{load} + V_{internal\ resistance}$
$R_s = R_1 + R_2 + \dots$	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$
$P = \frac{W}{t} \quad \text{or} \quad W = Pt$	
$W = VIt \quad \text{or} \quad W = I^2 R t \quad \text{or} \quad W = \frac{V^2}{R} t$	
$P = VI \quad \text{or} \quad P = I^2 R \quad \text{or} \quad P = \frac{V^2}{R}$	

ELECTRODYNAMICS

$\Phi = BA \cos \theta$	$emf = -\frac{N \Delta \Phi}{\Delta t}$	$F = IB \ell \sin \theta$
$V_p I_p = V_s I_s$	$\frac{N_s}{N_p} = \frac{V_s}{V_p}$	

PHOTONS AND ELECTRONS

$c = f \lambda$		$E = hf$ or $E = \frac{hc}{\lambda}$	
$E = W_0 + E_{K(max)}$	$W_0 = hf_0$	$E_{K(max)} = \frac{1}{2}mv_{max}^2$	